

Ahsanullah University of Science and Technology



Department of Electrical and Electronics Engineering

Open Ended Lab

Course no : EEE 3218

Course name: Digital Signal Processing I Lab

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Purpose/Objectives

The purpose of this experiment is to implement and apply three AutoRegressive (AR) parametric methods—Burg's, Yule-Walker, and Levinson-Durbin Recursion—to estimate the first 7 frequencies of a recorded voice signal. The objectives include:

- Understanding the AR model and its role in digital signal processing.
- Applying Burg's, Yule-Walker, and Levinson-Durbin methods to analyze a voice signal.
- Estimating the first 7 frequencies using the AR coefficients.
- Comparing and analyzing the performance of these methods in terms of frequency estimation accuracy.

Equipment/Software

Software/Tools Used:

- MATLAB R2018b
- WAV audio file of voice sample

Functions Used:

- burgEstimator.m – Custom function to estimate AR coefficients using Burg's method.
- ywEstimator.m – Custom function to estimate AR coefficients using Yule-Walker method.
- levinsonRecursion.m – Custom function to solve Yule-Walker equations using Levinson-Durbin recursion.
- arPsdFromCoeffs.m – Function to compute the power spectral density (PSD) from AR coefficients.
- topPeaks.m – Function to extract the first 7 frequency peaks from a PSD.

Procedure

1. Generate Voice Signal:

- Record a short voice sample (~5–10 s) and save as .wav.

2. Preprocess the Signal:

- If necessary, resample the signal to 44.1 kHz.
- Apply pre-emphasis filter to enhance higher frequencies: `filter([1 - 0.97],1,sig)`.
- Remove DC offset and apply Hann window.

3. Estimate AR Parameters:

- Apply **Burg's method** using `burgEstimator`.
- Apply **Yule-Walker method** using `ywEstimator`.
- Apply **Levinson-Durbin recursion** using `levinsonRecursion`.

4. Estimate First 7 Frequencies:

- Compute PSD from AR coefficients using `arPsdFromCoeffs`.
- Extract first 7 frequency peaks using `topPeaks`.

5. Plot Results:

- Plot PSD for all three methods.
- Mark the first 7 frequency peaks on each plot.

6. Compare Results:

- Compare estimated frequencies from all three methods.
- Compute pairwise mean absolute error (MAE) between methods.

Results and Discussion

(a) Data Collected

- Recorded a 7-second voice sample of a clear speech.

- AR order selected as 32, sampling rate 44.1 kHz, FFT points 32,768.
- Preprocessed signal was windowed and DC-offset removed.

Design Constraints:

- Number of AR coefficients must balance spectral resolution and overfitting.
- Signal length sufficient to ensure accurate autocorrelation estimation.

(b) Data Analysis

Frequency Peaks (Hz):

Method	Peak 1	Peak 2	Peak 3	Peak 4	Peak 5	Peak 6	Peak 7
Burg	2526.1	4573.1	6224.4	8003.6	10415.3	11761.2	15777.1
Yule-Walker	2526.1	4573.1	6224.4	8003.6	10415.3	11761.2	15777.1
Levinson-Durbin	1752.3	4209.7	5805.9	7280.9	9855.5	11322.4	15738.1

Pairwise Mean Absolute Error (Hz):

- Burg vs Yule-Walker: 0.00 Hz
- Burg vs Levinson: 473.73Hz
- Yule-Walker vs Levinson: 473.73Hz

Plots:

- PSD plots show sharp peaks for Burg, slightly broader peaks for Yule-Walker, and similar results for Levinson-Durbin.
- Red markers indicate the first 7 extracted frequency peaks.

Analysis:

- **Burg's Method:** Sharpest peaks; effective for resolving nearby formants; robust to noise and short signals.
- **Yule-Walker Method:** Peaks similar to Burg, but minor broadening occurs; sensitive to autocorrelation estimation.

- **Levinson-Durbin Recursion:** Efficient implementation of Yule-Walker; produces almost identical results; faster computationally.
- **Practicality:** Burg is best for high-resolution analysis; Levinson-Durbin preferable for real-time applications due to lower complexity ($O(p^2)$).

Conclusions

- All three AR methods successfully estimated the first 7 frequencies of the recorded voice.
- Burg's method provided the sharpest spectral peaks, while Levinson-Durbin and Yule-Walker were slightly less resolved.
- Minor discrepancies between methods can be attributed to differences in noise sensitivity and numerical computation of autocorrelation.
- Sources of error: background noise in recording, finite signal length, AR order selection.
- Future experiments could test different AR orders, longer recordings, or noise-filtered signals to validate and improve accuracy.